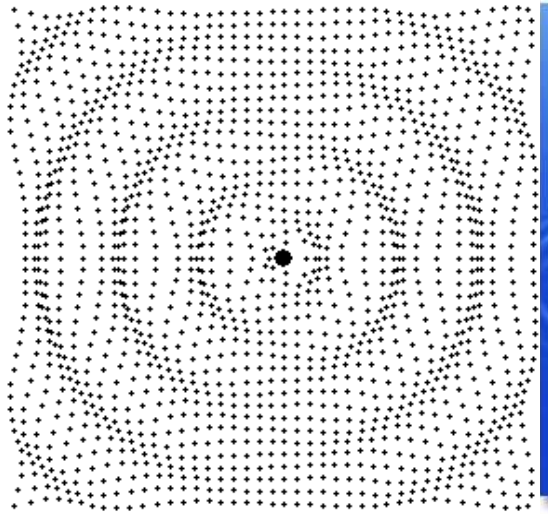
Vibration & Waves

CHAPTER 3

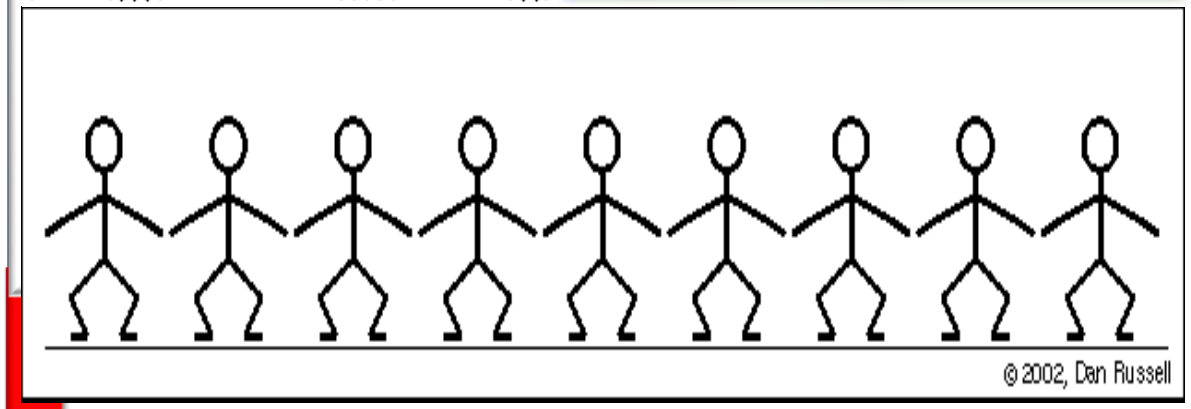
Waves

Waves

Is a disturbance that propagates through space and time, usually with transference of energy.







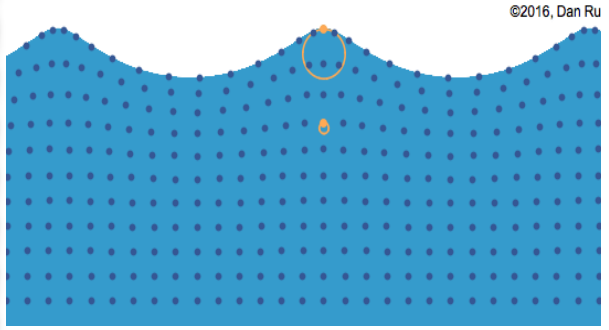
Waves

Mechanical Waves

Propagate through a medium

Examples:

sound; water waves;
"people wave"
in a football stadium

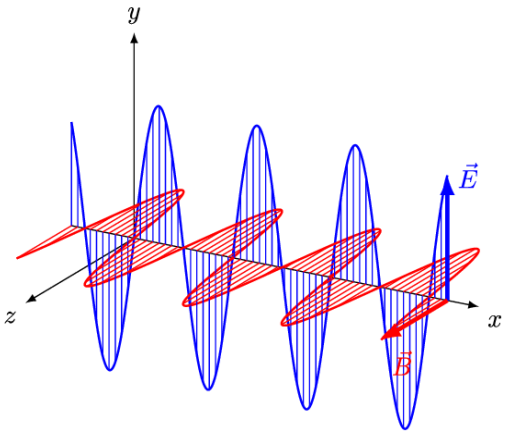


Electromagnetic Waves

Propagate through space

Examples:

Light, TV and radio waves

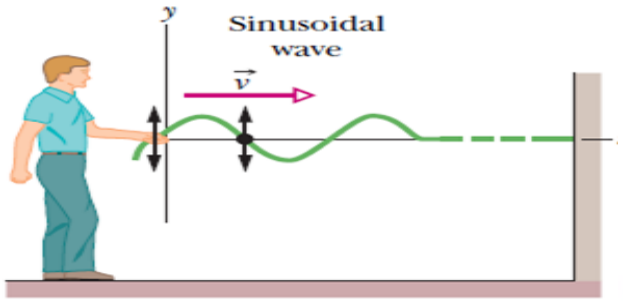


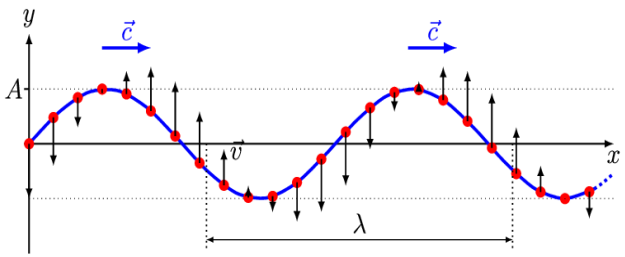
Waves

Transverse Waves

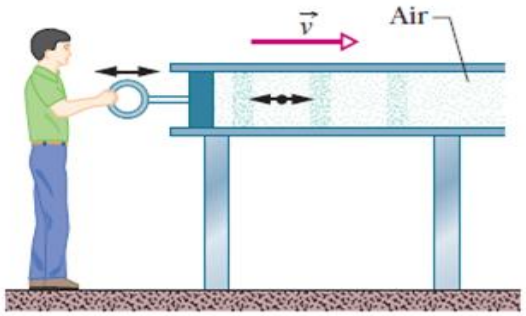
Longitudinal Waves

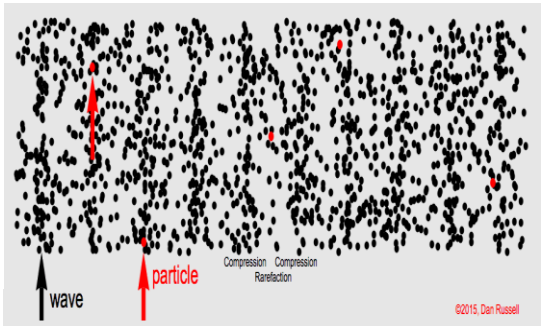
Waves on a string



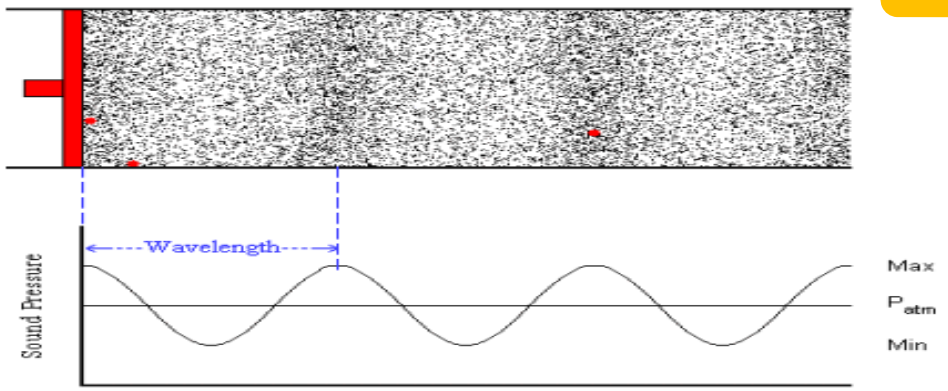


Sound waves





Important Concepts:



Wavelength: Is the distance between two successive points in the wave that have the same phase.

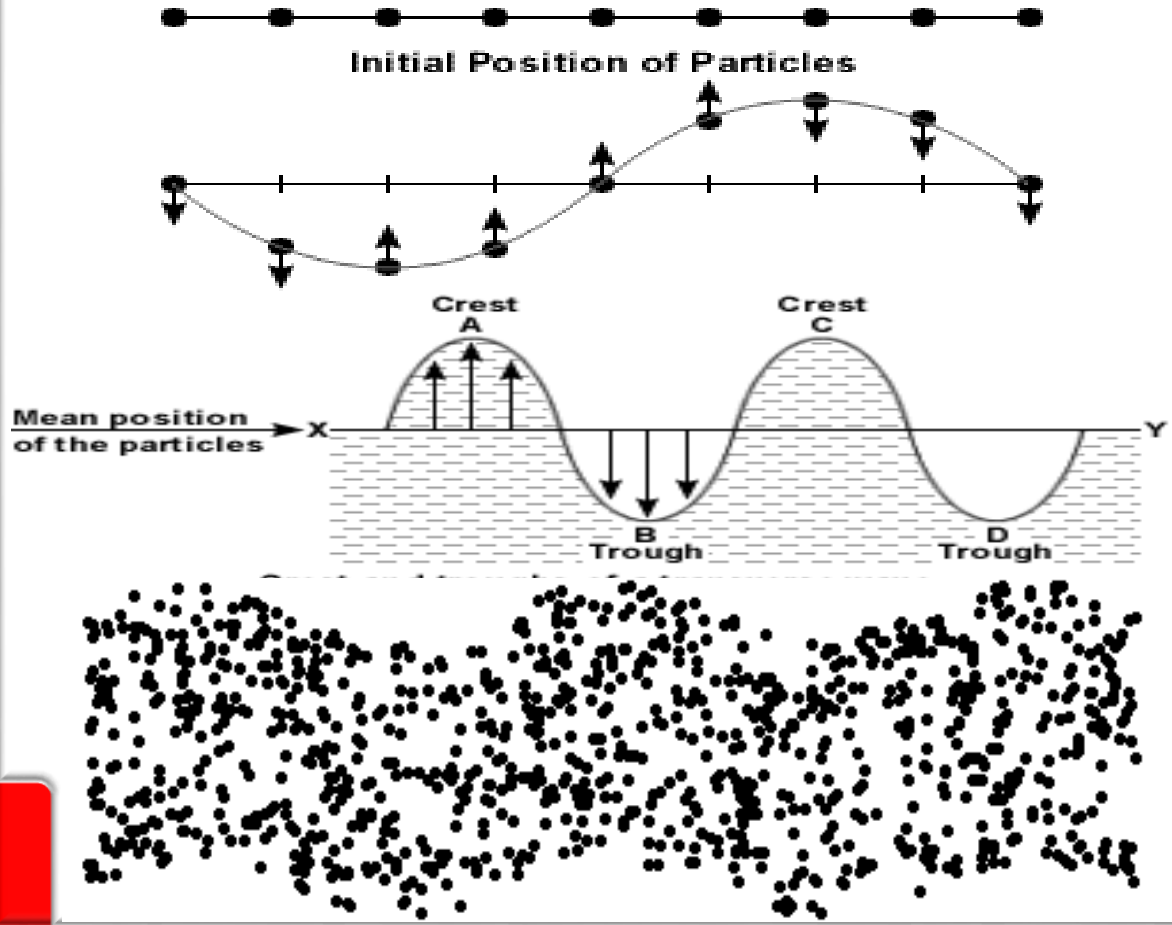
Frequency: Is the number of waves per unit time.

Periodic time: Is the reciprocal of the freq., Or/Is the time needed to make one complete wave.

Amplitude: Is the maximum displacement in the direction of vibration.

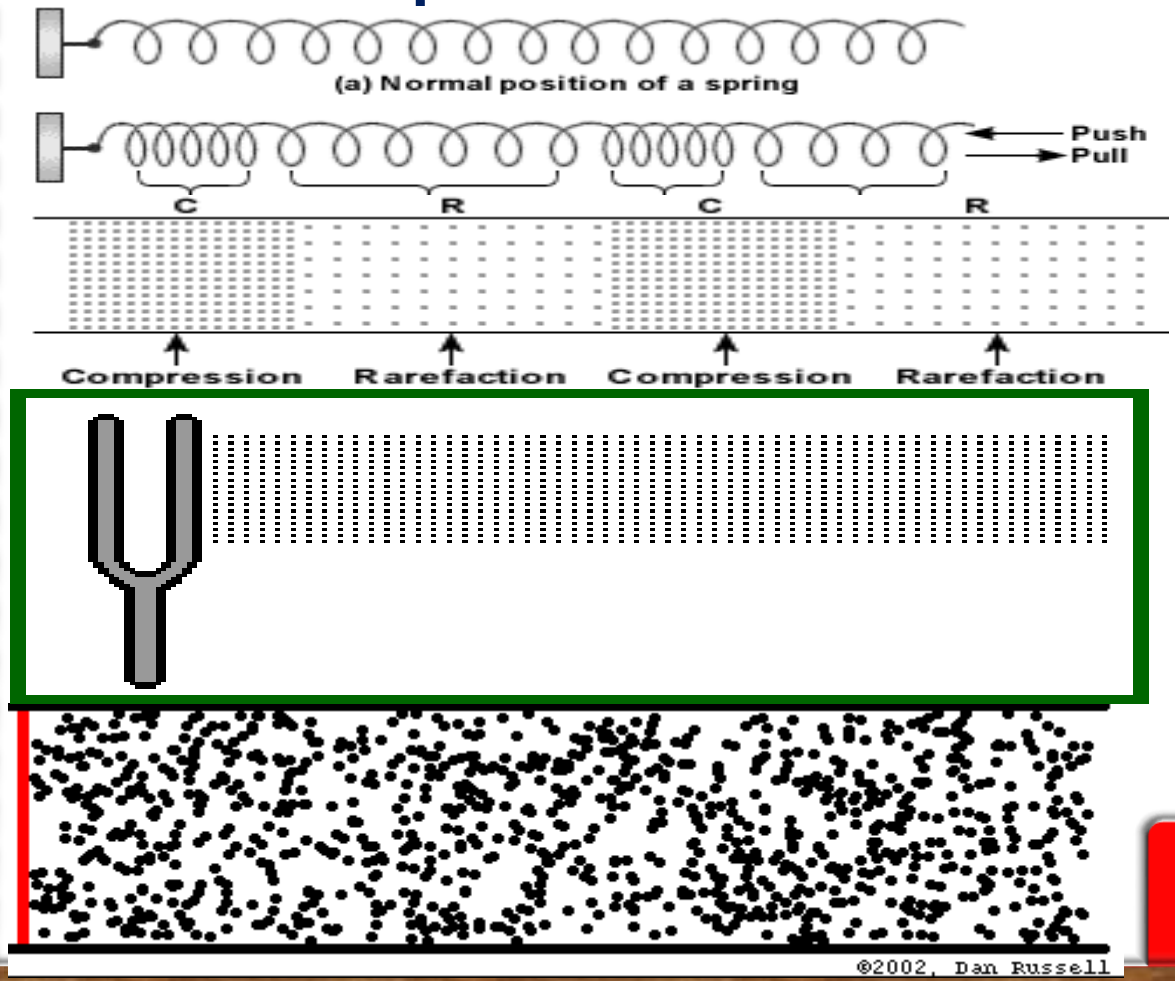
Transverse Waves

The particles of the disturbed medium vibrate perpendicular to the wave propagation and form crests and troughs



Longitudinal Waves

The particles of the disturbed medium vibrate parallel to the wave propagation and form compressions and rarefactions



Wave Equation

Waves propagate from one place to another.

The Displacement equation:

$$Y_{(x,t)} = y_m \sin(kx - \omega t)$$

Where: $Y_{(x,t)}$ is the displacement, y_m is an amplitude, x is a position, t is time, $\sin(kx-\omega t)$ is oscillating term, $(kx-\omega t)$ is the phase angle, k is an angular wavenumber and ω is an angular frequency.

Transverse Velocity of Vibrating String

$$\because y(x,t) = y_m \sin(kx - \omega t)$$

$$\text{Speed } (v) = \frac{dy}{dt} = -\omega y_m \cos(kx - \omega t)$$

$$v_{\max} = \omega y_m$$

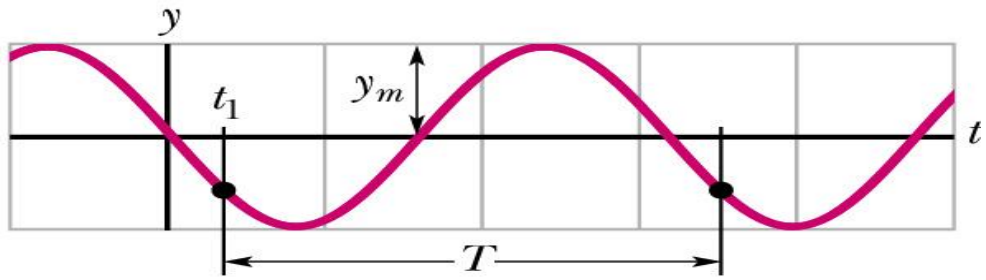
Transverse Acceleration of Vibrating String

$$(a) = \frac{dv}{dt} = \frac{d(-\omega y_m \cos(kx - \omega t))}{dt}$$

$$a = -\omega^2 y_m \sin(kx - \omega t)$$

$$a_{\max} = \omega^2 y_m$$

The relation between periodic time and frequency



If position is fixed, then x =0

$$y(0,t) = y_m \sin(-\omega t) = -y_m \sin(\omega t)$$
$$= -y_m \sin(\omega t + \omega T)$$

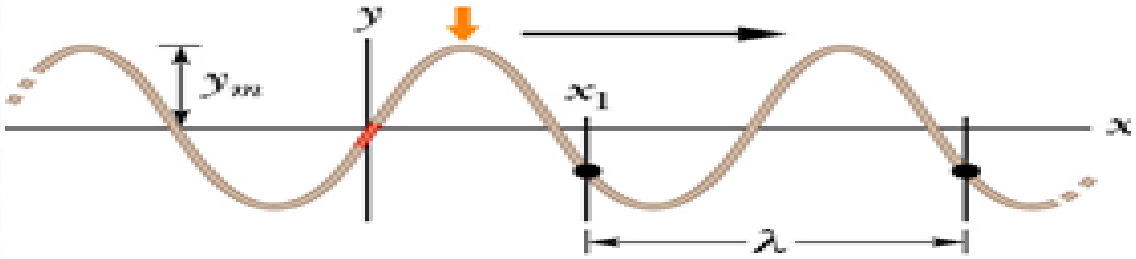
The sin function repeats itself by increasing angle 2π

$$\therefore \omega T = 2\pi \qquad T = \frac{2\pi}{\omega} = \text{Periodic Time}$$

$$\omega = 2\pi f = \text{angular frequency}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{T} = \text{frequency}$$

The relation between wavelength and wavenumber



If time is fixed, then t=0

$$y(x,0) = y_m \sin(kx) = y_m \sin(kx + k\lambda)$$

The sin function repeats itself by increasing angle 2π

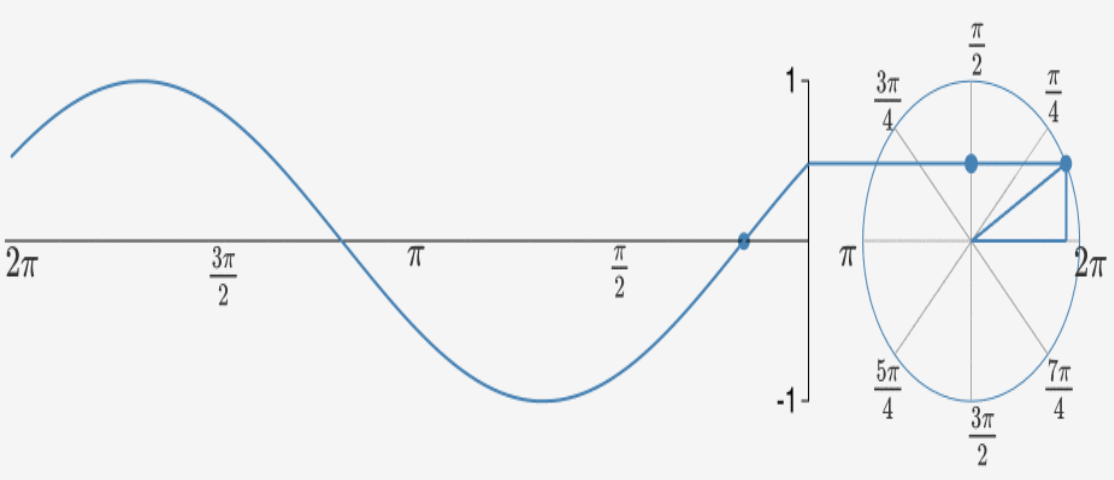
$$\therefore k\lambda = 2\pi \qquad \lambda = \frac{2\pi}{k} = \text{wavelength}$$

$$k = \frac{2\pi}{\lambda} = \text{angular wave number}$$

$$K = \frac{1}{\lambda} = \text{wave number}$$

Speed of traveling Wave

Is the distance covered by the wave in one second in a certain direction.



$$v = \lambda \times f = \frac{2\pi}{2\pi} f \times \lambda = 2\pi f \times \frac{\lambda}{2\pi} = \frac{\omega}{k}$$

$$v = \lambda \times f \qquad \therefore v = \omega / k$$

Example:

Sinusoidal wave traveling along a string is described By $Y(x,t) = 0.00327\, sin[\,(72.1(x) - 2.72t)]$ **What is the amplitude of this wave? What are the wavelength and the period of this wave? What are the wave number and the frequency of this wave ? What is the speed of this wave?**

$$Y(x,t) = y_m\, sin(kx - \omega t)$$

$$y_m = 0.00327\, m$$

$$k = 72.1\ \mathrm{rad/m} \qquad \lambda = 2\pi / k = 0.0871 m$$

$$\omega = 2.72\, rad/s \qquad T = 2\pi/\omega = 2.31\, s$$

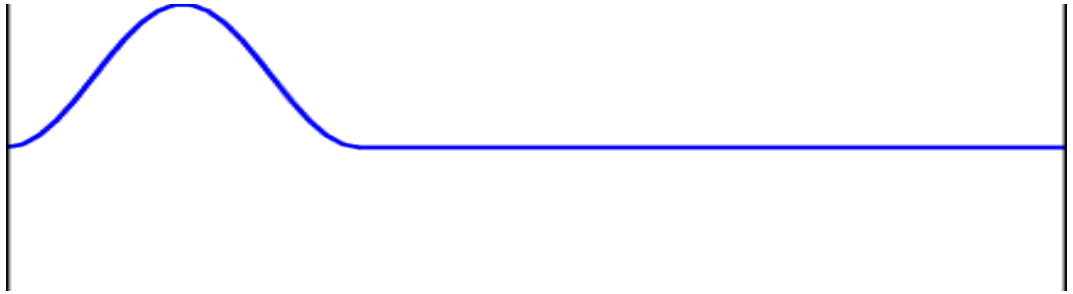
$$K = 1/\lambda = 11.5\, m^{-1}$$

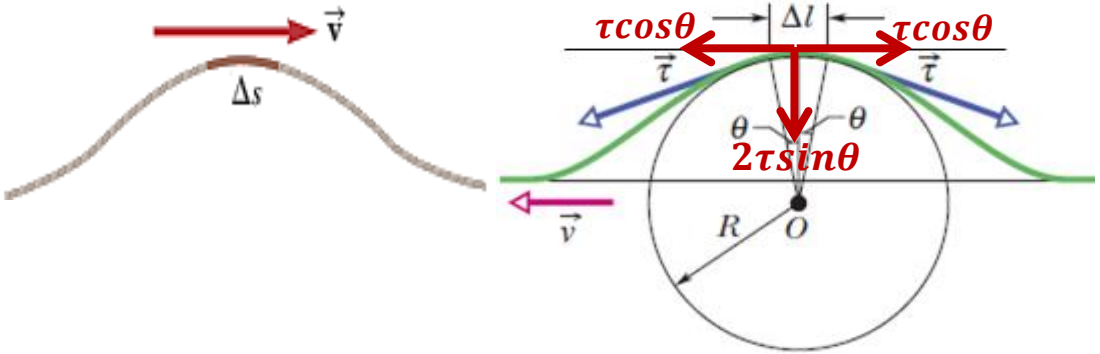
$$f = 1/T = 0.433\, Hz$$

$$v = \lambda f = 0.377\, m/s$$

Speed of Waves on Stretched Strings

Let's consider a pulse moving to right and look at it in the frame that moves along with the pulse.





$$\therefore \sum F_x = \tau cos\theta - \tau cos\theta = zero$$

$$\sum F_y = 2\tau sin\theta \qquad \approx 2\tau\theta = \tau(\Delta l/R)$$

$$, \because \sum F = ma$$

Since in the reference frame moves with the pulse, the segment is moving to the left with the speed v, and the centripetal acceleration of the segment is:

$$a_r = \frac{v^2}{R}$$

What is the mass of the segment when the line density of the string is m?

$$m = \mu\Delta l$$

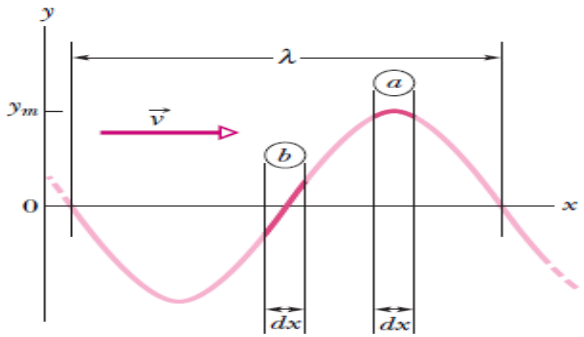
$$\therefore \sum F = ma = \mu\Delta l\frac{v^2}{R}$$

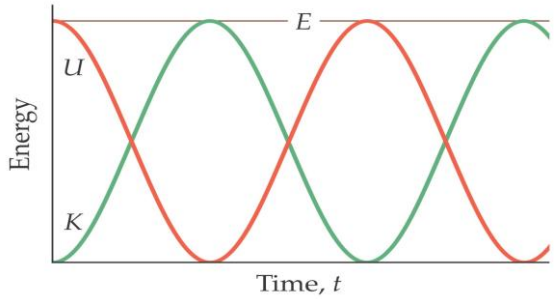
Therefore, $$\mu\Delta l\frac{v^2}{R} = \tau(\frac{\Delta l}{R})$$

$$\therefore v = \sqrt{\frac{\tau}{\mu}}$$

Energy and Power

transfer by sinusoidal waves on strings





$$\boldsymbol{E = K.E + P.E} \qquad \because \boldsymbol{Power = \frac{Energy}{time}}$$

$$\therefore P_{avg} = (\frac{dK}{dt})_{avg} + (\frac{dU}{dt})_{avg}$$

$$\because (\frac{dK}{dt})_{avg} = (\frac{dU}{dt})_{avg}$$

$$\therefore \boldsymbol{P_{avg} = 2(\frac{dK}{dt})_{avg}}$$

$$\because K.E = \frac{1}{2}mv^2 \qquad \because m = \mu l$$

$$\because v^2 = w^2 y_m^2 \cos^2(kx - wt)$$

$$\therefore K.E = \frac{1}{2}(ul)(w^2 y_m^2 \cos^2(kx - wt)$$

$$\frac{dK}{dt} = \frac{1}{2}uvw^2 y_m^2 \cos^2(kx - wt)$$

$$(\frac{dK}{dt})_{avg} = \frac{1}{2}uvw^2 y_m^2 (\cos^2(kx - wt))_{avg}$$

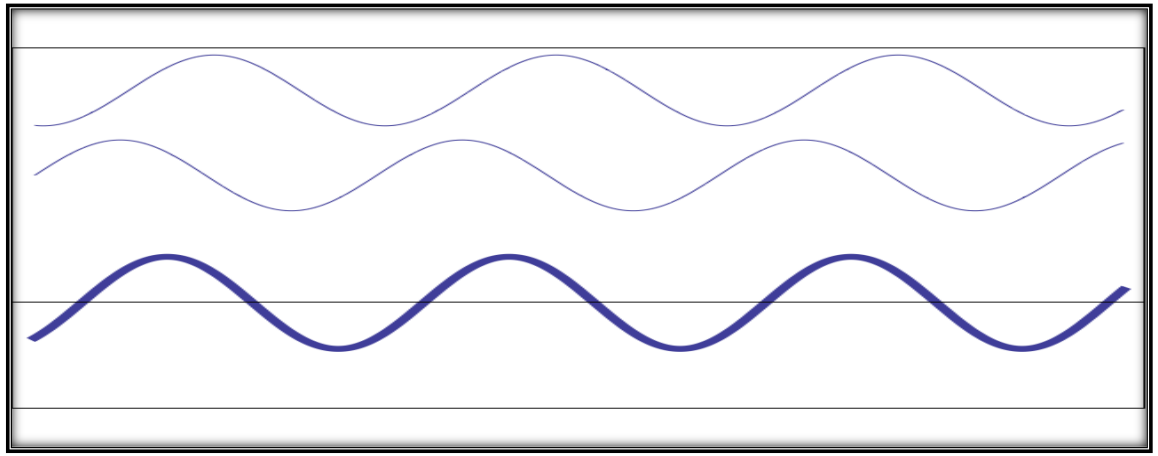
$$\because (\cos^2(kx - wt))_{avg} = \frac{1}{2}$$

$$\therefore (\frac{\boldsymbol{dK}}{\boldsymbol{dt}})_{avg} = \frac{\boldsymbol{1}}{\boldsymbol{4}}\boldsymbol{uvw^2 y_m^2}$$

$$\therefore \boldsymbol{P_{avg} = \frac{1}{2}uvw^2 y_m^2}$$

Interference of Waves

Is the overlap of two or more waves to produce a new wave has a new intensity.



If two sinusoidal waves of the same amplitude and wavelength travel in the same direction along a stretched string, they interfere to produce a resultant sinusoidal wave traveling in that direction.

$$Y_{(x,t)} = Y_{1(x,t)} + Y_{2(x,t)}$$

$$y_m \sin(kx - \omega t) \qquad y_m \sin(kx - \omega t + \varphi)$$

$$\therefore Y_{(x,t)} = y_m \sin(kx - \omega t) + y_m \sin(kx - \omega t + \varphi)$$

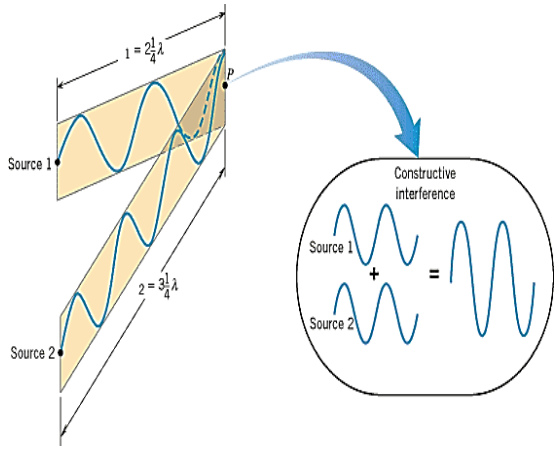
$$\sin\alpha + \sin\beta = 2\cos\frac{1}{2}(\alpha - \beta)\sin\frac{1}{2}(\alpha + \beta)$$

$$\alpha = kx - \omega t \qquad \beta = kx - \omega t + \varphi$$

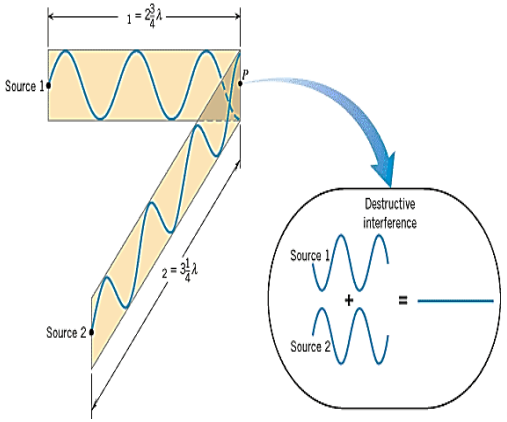
$$y(x,t) = 2y_m \cos\frac{1}{2}\varphi \sin(kx - \omega t + \frac{1}{2}\varphi)$$

$$Amplitude = 2y_m \cos\frac{1}{2}\varphi$$

Constructive interference



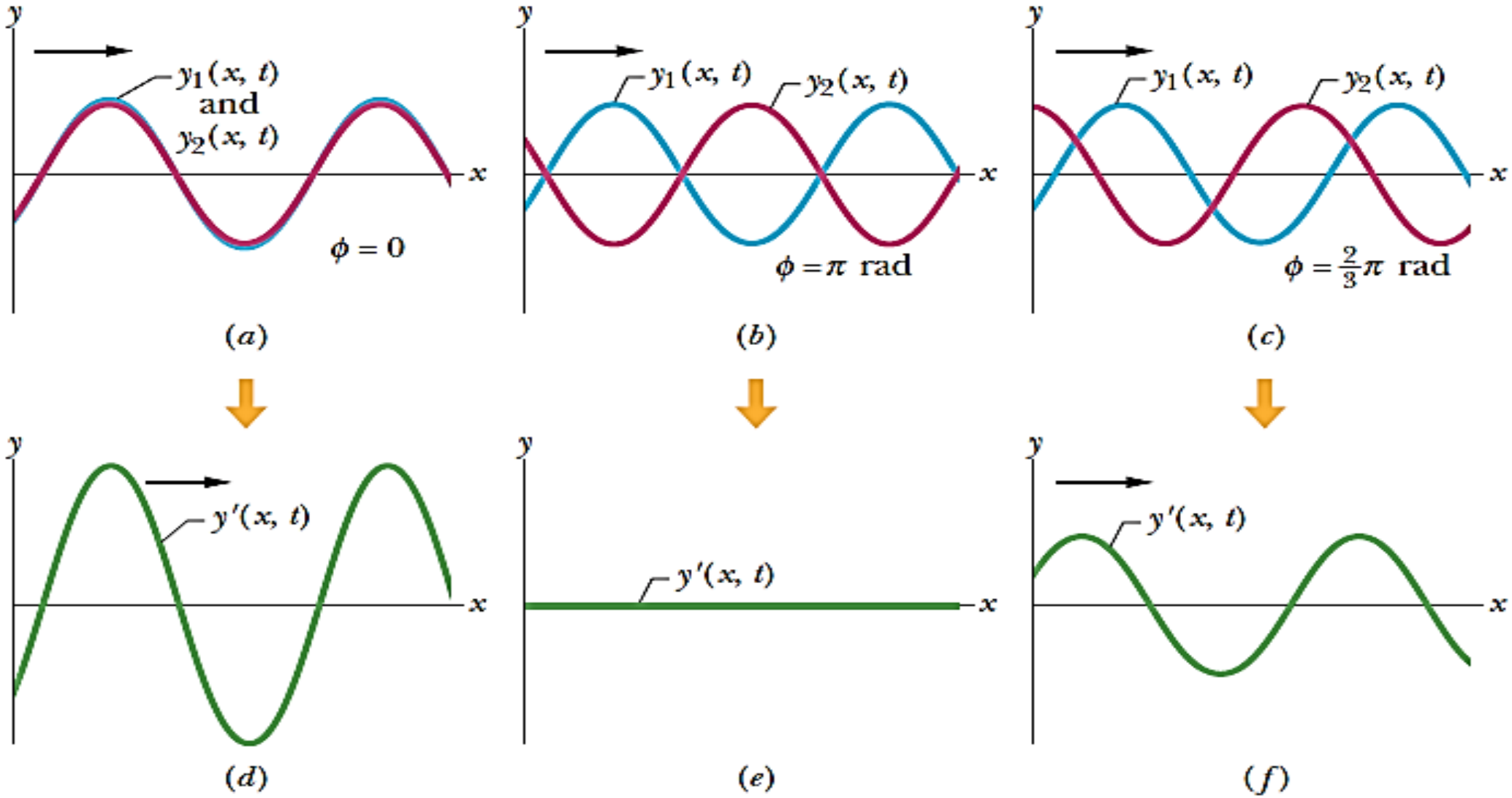
Destructive interference



Being exactly in phase, the waves produce a large resultant wave.

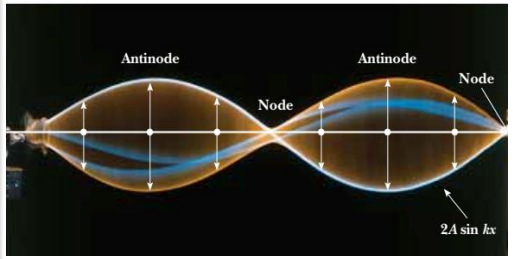
Being exactly out of phase, they produce a flat string.

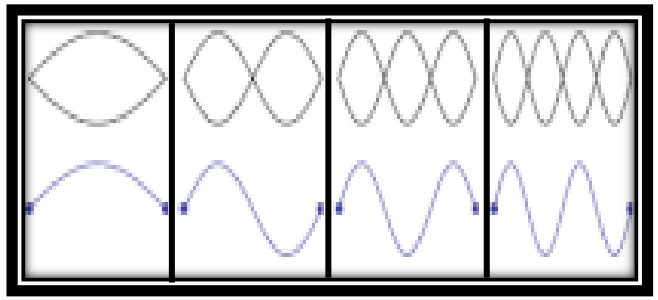
This is an intermediate situation, with an intermediate result.



Standing Waves

The interference of two identical sinusoidal waves moving in opposite directions





If two sinusoidal waves of the same amplitude and wavelength travel in opposite directions along a stretched string, their interference with each other produces a standing wave.

$$y_{1(x,t)} = y_m \sin(kx - \omega t)$$

$$y_{2(x,t)} = y_m \sin(kx + \omega t)$$

$$\therefore y_{(x,t)} = y_{1(x,t)} + y_{2(x,t)}$$

$$\therefore y_{(x,t)} = y_m[\sin(kx - \omega t) + \sin(kx + \omega t)]$$

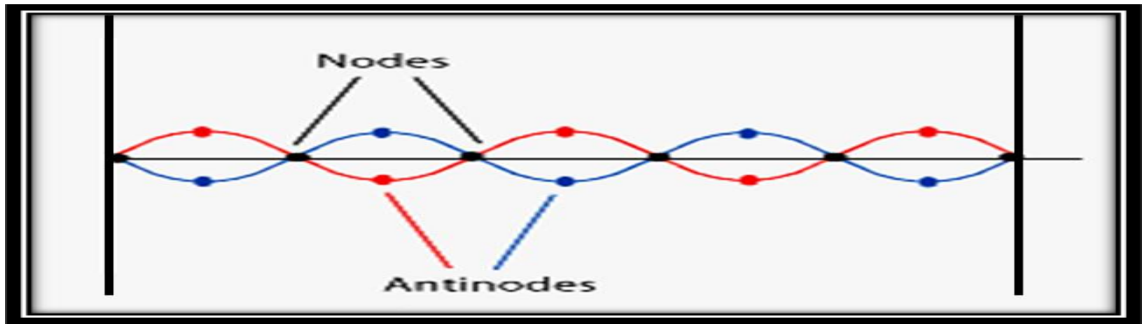
$$\sin\alpha + \sin\beta = 2\cos\frac{1}{2}(\alpha - \beta)\sin\frac{1}{2}(\alpha + \beta)$$

$$\alpha = kx - \omega t$$

$$\beta = kx + \omega t$$

$$y_{(x,t)} = [2y_m \sin k\,x](\cos \omega\, t)$$

$$Amplitude = 2y_m \sin k\,x$$

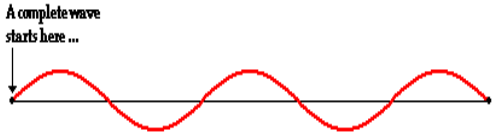


Nodes: is the positions of zero amplitude in standing wave eq. $x = n\lambda/2$, *for n =0,1,2, . . .*

Antinodes: is the positions of maximum amplitude in standing wave eq.

$x = (n+1/2)\,\lambda/2$, *for n 0,1,2, . . .*

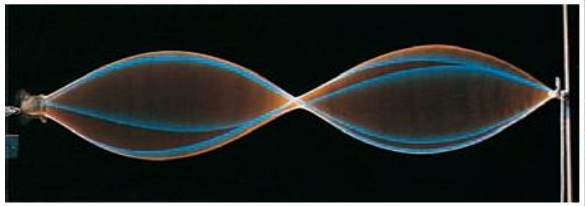
Standing Waves and Resonance

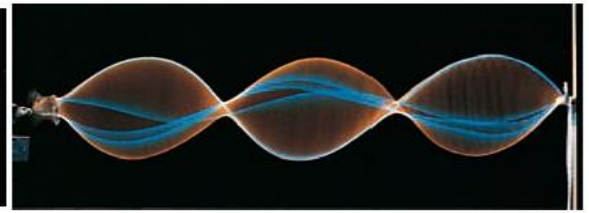


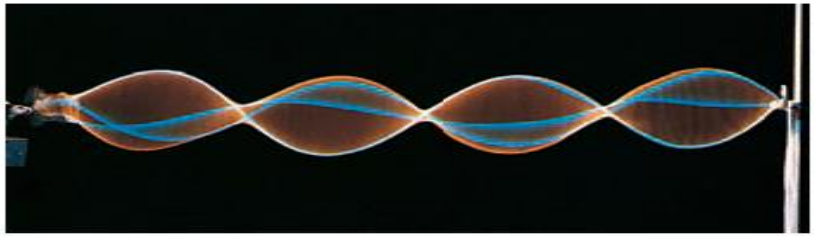


For certain frequencies, the interference produces a standing wave pattern (or oscillation mode) with nodes and large antinodes like those in Fig.

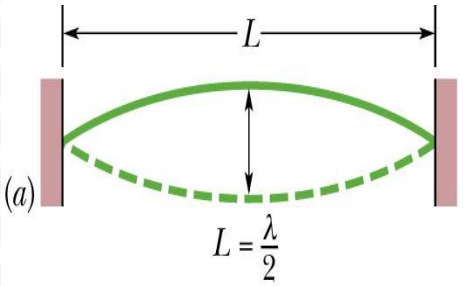
Such a standing wave is said to be produced at resonance, and the string is said to resonate at these certain frequencies, called resonant frequencies.



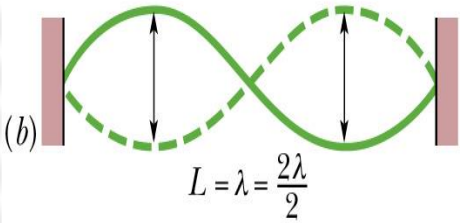




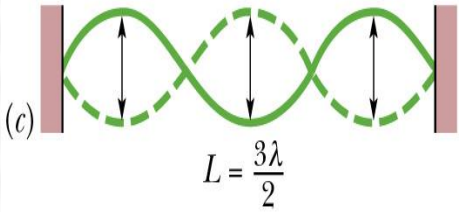
Resonant Frequencies



Fundamental or first harmonic $f = \frac{v}{2l}$



Second harmonic $f = \frac{2v}{2l}$



Third harmonic $f = \frac{3v}{2l}$

$\because v = \lambda f$

$$\therefore f = \frac{v}{\lambda} = \frac{nv}{2l}$$

